Lecture 3: Enzymes and Metabolism

Enzymes are biological catalysts that speed up chemical reactions in living cells without being consumed. Most enzymes are proteins, and each one has an active site that binds a specific substrate.

The rate of an enzyme-catalysed reaction depends on temperature, pH, and the concentration of the substrate. Above the optimum temperature, the enzyme denatures and loses its shape.

Key properties

- Enzymes lower the activation energy of a reaction.
- They are specific to their substrate.
- They can be inhibited by competitive and non-competitive inhibitors.

Michaelis–Menten kinetics

The Michaelis–Menten equation describes how the reaction velocity depends on substrate concentration:

$$v = V_{max} \times [S] / (K_m + [S])$$

Here Km is the substrate concentration at which the velocity is half of Vmax. A low Km means high affinity between enzyme and substrate.

Enzyme	Substrate	Km (mM)
Hexokinase	Glucose	0.15
Catalase	H2O2	25

The following code estimates the velocity:

```
def velocity(vmax, km, s):
    return vmax * s / (km + s)
```

الإنزيمات في الجسم

الإنزيمات بروتينات تسرّع التفاعلات الكيميائية في الخلايا الحية دون أن تُستهلك. ولكل إنزيم موقع نشط يرتبط بمادة متفاعلة محددة.

تتأثر سرعة التفاعل الإنزيمي بدرجة الحرارة ودرجة الحموضة وتركيز المادة المتفاعلة.

Scanned page: Competitive inhibitors bind to the active site of the enzyme.

Summary

Enzymes make metabolism possible. Remember the Michaelis–Menten equation and the meaning of Km for the exam.